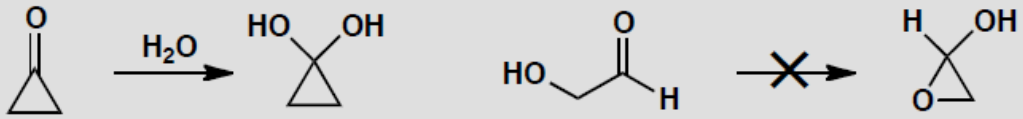
Tutorial 6

Problem 1

Cyclopropanone exists as the hydrate in water but 2-hydroxyethanal does not exist as the hemiacetal. Explain.

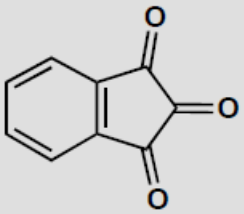


Hydration is an equilibrium reaction so the mechanism is not strictly relevant to the question, though there is no shame in including mechanisms whenever you can. To answer the question we must consider the effect of the three-membered ring on the relative stability of starting material and product. All three-membered rings are very strained because the bond angles are 60° instead of 109° or 120°. Cyclopropanone is particularly strained because the sp2 carbonyl carbon would like a bond angle of 120°— there is '60° of strain.' In the hydrate that carbon atom is sp3 hybridized and so there is only about '49° of strain.' Not much gain, but the hydrate is more stable than the ketone.

The second case is totally different. The hydroxy-aldehyde is not strained at all but the hemiacetal has '49° of strain' at each atom. Even without strain, hydrates and hemiacetals are usually less stable than their aldehydes or ketones because one C=O bond is worth more than two C–O bonds. In this case the hemiacetal is even less stable and, unlike the cyclopropanone, can escape strain by breaking a C–O ring bond.

Problem 2

The triketone shown here is called 'ninhydrin' and is used for the detection of amino acids. It exists in aqueous solution as a hydrate. Which ketone is hydrated and why?

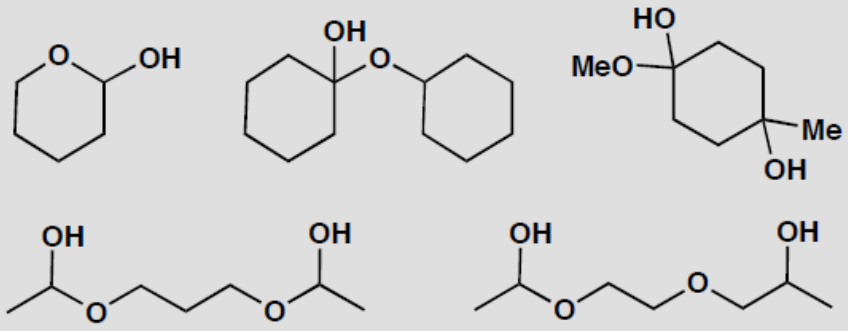


The two ketones next to the benzene ring are stabilized by conjugation with it but also destabilized by the central ketone—two electron-withdrawing groups next to each other is a bad thing. The central carbonyl group is not stabilized by conjugation and is destabilized by

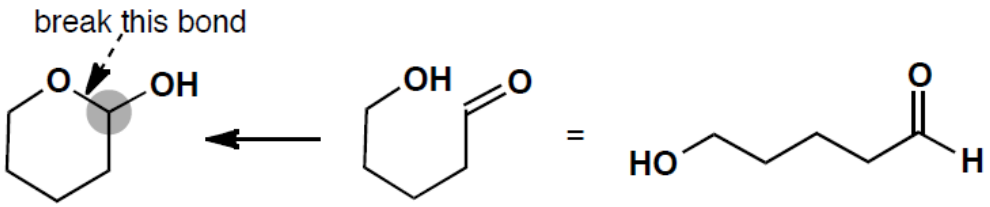
two other ketones so it forms the hydrate. Did you remember that hydrate formation is thermodynamically controlled?

Problem 3

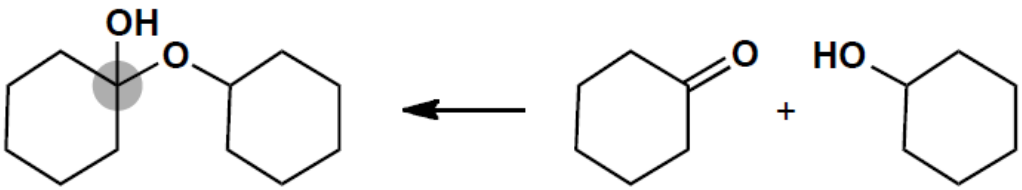
Each of these compounds is a hemiacetal and therefore formed from an alcohol and a carbonyl compound. In each case give the structures of the original materials.



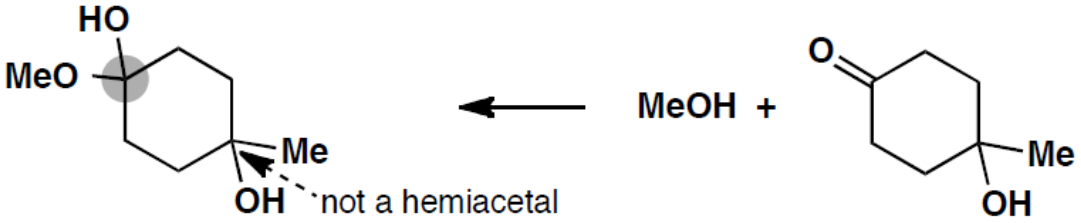
Each OH group represents a carbonyl group in disguise (marked with a grey circle). Just break the bond between this carbon and the other oxygen atom and you will see what the hemiacetal was made from. The first example shows how it is done.



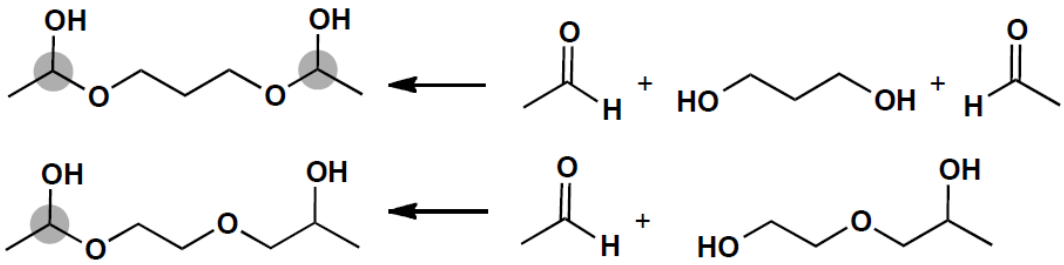
The next is similar but the alcohol is a different molecule.



Do not be deceived by the third example. There is one hemiacetal (two oxygens joined to the same carbon atom) but the other OH is just a tertiary alcohol.

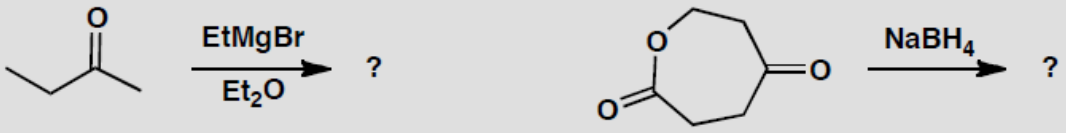


The last two examples are not quite the same. The first is indeed symmetrical but the second has one oxygen atom in a different position so that there is only one hemiacetal. Note that these hemiacetals may not be stable.

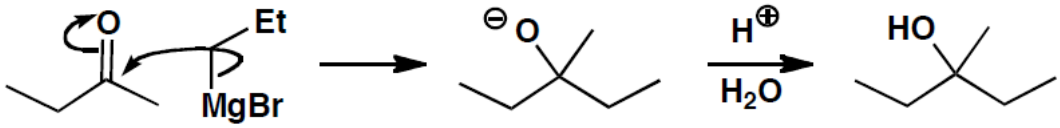


Problem 4

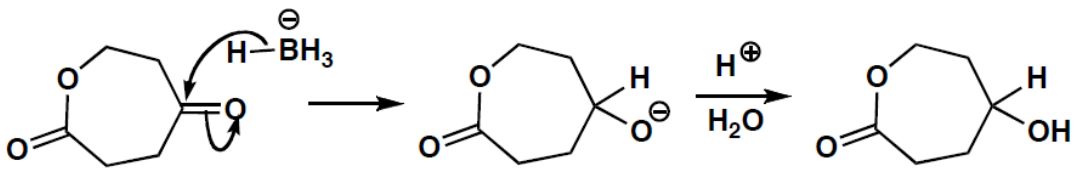
What would be the products of these reactions? In each case give a mechanism to justify your prediction.



The Grignard reagent will add to the carbonyl group and the work-up will give a tertiary alcohol as the final product.



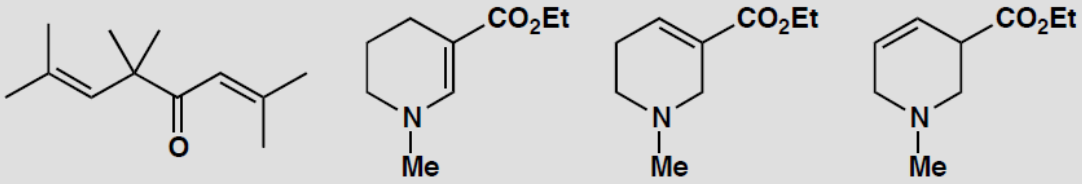
The second reaction should give you brief pause for thought as you need to recall that borohydride reduces ketones but not esters.



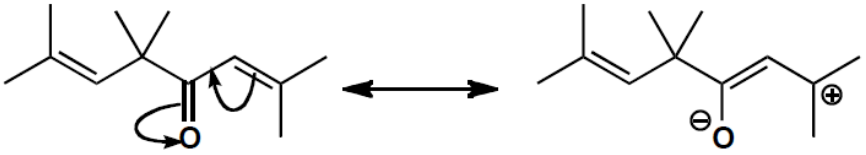
Tutorial 7

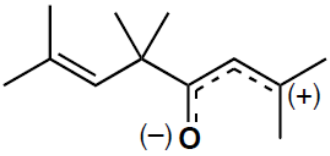
Problem 1

Are these molecules conjugated? Explain your answer in any reasonable way.

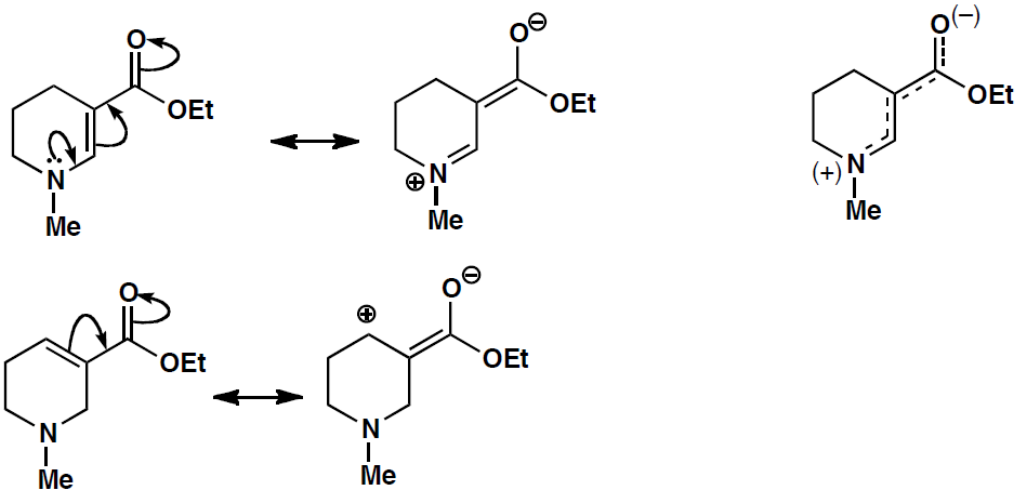


The first compound is straightforward with one conjugated system (an enone) and a non-conjugated alkene. You could draw curly arrows to show the conjugation, like this, and/or give a diagram to show the distribution of the electrons.

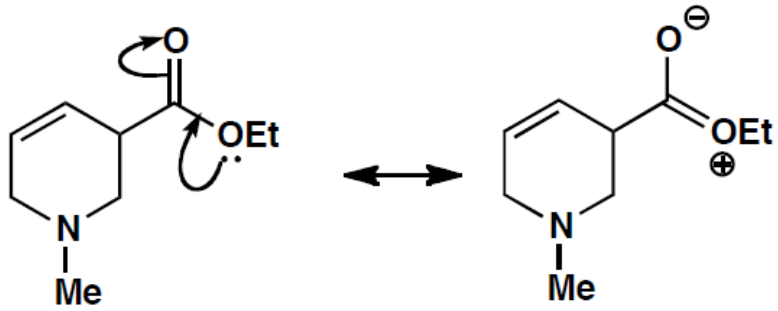




The last three compounds obviously form a related group with the same skeleton and only the alkene moved round. There is of course ester conjugation in all three and this is the only conjugation in the last molecule. The first has extended conjugation between the nitrogen lone pair and the carbonyl group and the second has simple conjugation between the alkene and the ester.



The only conjugation in the last compound is the delocalization of the ester oxygen lone pair. This is of course there in all the other compounds too.

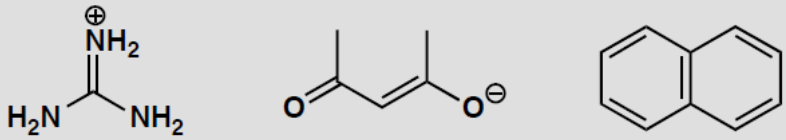


Problem 2

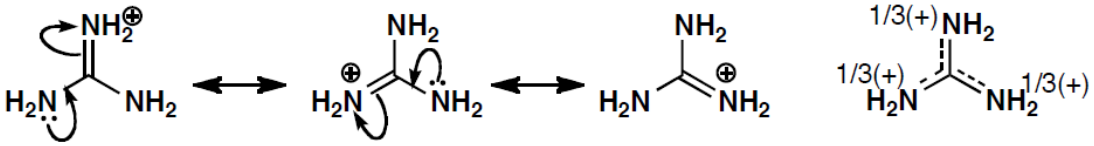
Draw diagrams to represent the conjugation in these molecules. Draw two types of diagram:

(a) Show curly arrows linking at least two different ways of representing the molecule

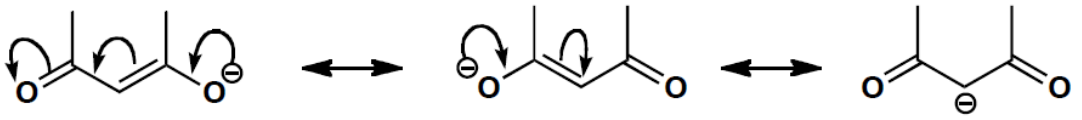
(b) Indicate with dotted lines and partial charges (where necessary) the partial double bond (and charge) distribution.

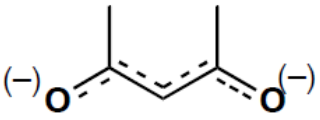


Treating each compound separately in the styles demanded by the question, the first (the guanidinium ion) is a very stable cation because of conjugation. The charge is delocalized onto all three nitrogen atoms as the first three structures show. Each nitrogen has an equal positive charge so our fourth diagram shows one third + on each.

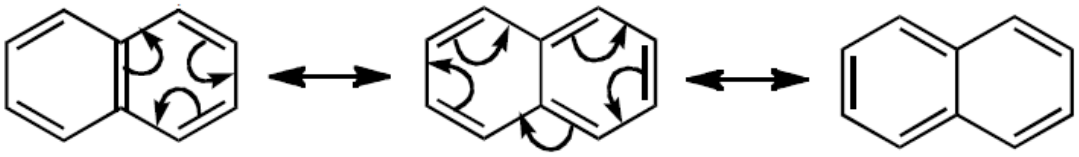


The second compound is what you will learn to call an enolate anion. The negative charge is delocalized throughout the molecule, mostly on the oxygens but some on carbon. It is difficult to represent this with partial charges but the charges on the oxygens will be nearly a half each.



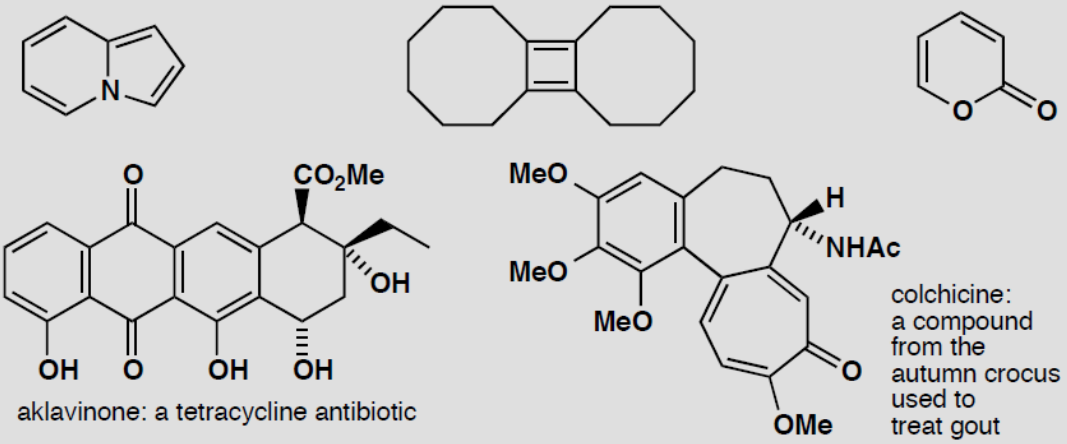


The third compound is naphthalene. The structure drawn in the question is the best as both rings are benzene rings. The results of curly arrow diagrams show how naphthalene is delocalized all round the outer ring. In fact these diagrams show the ten electrons in the outer ring – this is a 4n + 2 number and all three diagrams show that naphthalene is aromatic.



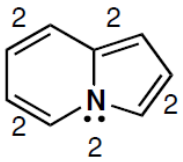
Problem 3

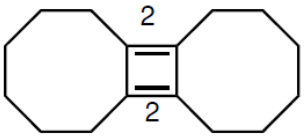
Which (parts) of these compounds are aromatic? Justify your answer with some electron counting. You may treat rings separately or together as you wish. You may notice that two of them are compounds we met in problem 2 of this chapter.

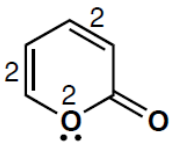


The numbers show how many π electrons there are in each bond or at each atom. The first compound has a lone pair on nitrogen in a p-orbital shared between both rings. Each ring has

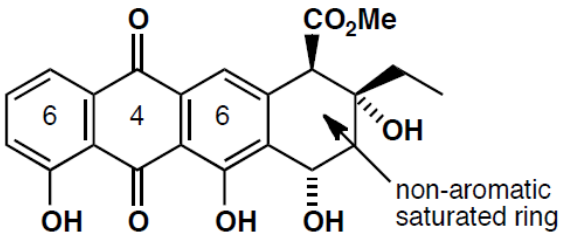
six electrons and the periphery of the whole molecule has ten electrons. Both rings and the entire molecule are aromatic. The second has four π electrons only so there is no aromaticity anywhere. The third has six π electrons in the ring including the lone pair on oxygen but not including the carbonyl group which is outside the ring. The compound is aromatic.

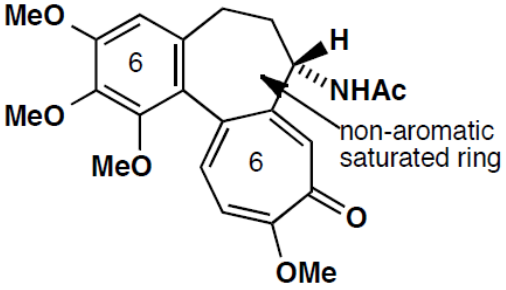






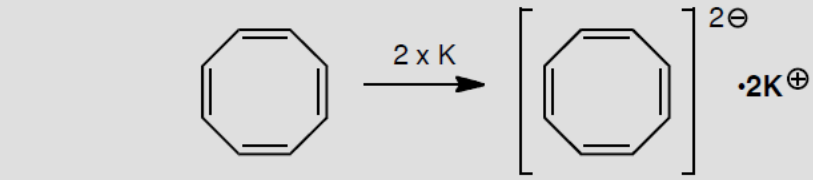
For the rest we have put the number of π electrons inside each ring and there are two aromatic rings in each compound. Again we don't count carbonyl group electrons as they are outside the ring. So one ring in aklavinone has only four electrons and is not aromatic while one of the seven-membered rings in colchicine is aromatic. Each compound has one saturated ring that cannot be aromatic.

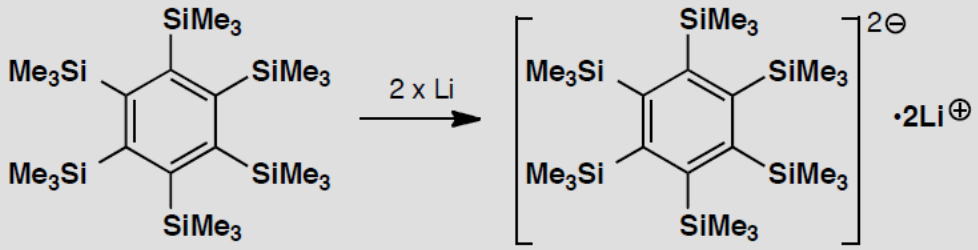




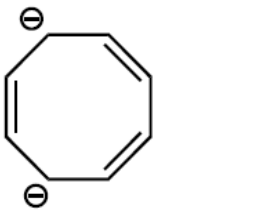
Problem 4

Cyclooctatetraene (see p. 158 of the textbook) reacts readily with potassium metal to form a salt, K_2[cyclooctatetraene]. What shape do you expect the ring to have in this compound? A similar reaction of hexa(trimethylsilyl)benzene with lithium also gives a salt. What shape do you expect this ring to have?

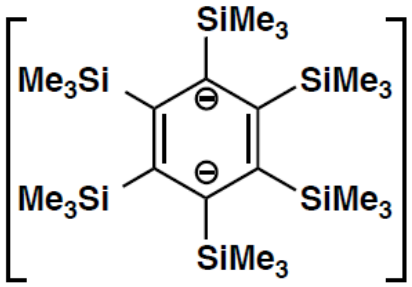




Cyclooctatetraene, as explained on p. 158 of the textbook, is 'tub-shaped' and not planar, because its eight π electrons do not form a 2n+2 number. However, two atoms of potassium can reduce the cyclooctatetraene to a dianion by giving it two electrons, so now it has ten electrons, is aromatic, and is planar. Just one of the many possible delocalized structures of the product is shown here. When lithium reduces hexa(trimethylsilyl)benzene, the aromatic sextet is increased to a total of eight delocalized electrons, so the compound is no longer aromatic. The six membered ring in the salt is no longer flat.



one way of drawing
the flat, aromatic dianion

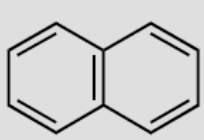


this dianion is no longer flat

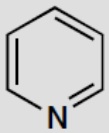
Tutorial 8

Problem 1

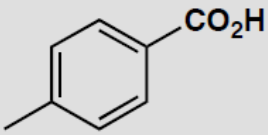
How would you separate a mixture of these three compounds?



naphthalene



pyridine

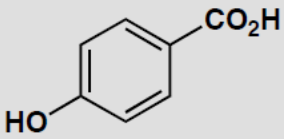


para-toluic acid

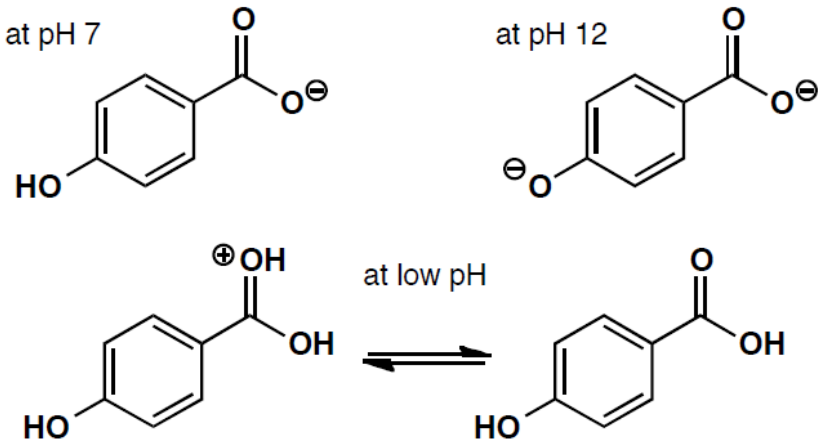
Pyridine is a weak base (pKa of the pyridinium ion is about 5.5) and can be dissolved in aqueous acid. Naphthalene is neither an acid nor a base and is not soluble in water at any pH. p-Toluic acid is a weak acid (pKa about 4.5) and can be dissolved in aqueous base. So dissolve the mixture in an organic solvent immiscible with water (say ether Et_2O or dichloromethane CH_2Cl_2) and extract with aqueous acid. This will dissolve the pyridine as its cation. Then extract the remaining organic layer with aqueous base such as $NaHCO_3$ which will remove the toluic acid as its water-soluble anion. You now have three solutions. Evaporate the organic solution to give crystalline naphthalene. Acidify the basic solution of p-toluic acid and the free acid will precipitate out and can be recrystallized. Add base to the pyridine solution, extract the pyridine with an organic solvent, and distil the pyridine. It doesn't matter if you extract the original solution with base first and acid second.

Problem 2

What species would be present in a solution of this hydroxy-acid in (a) water at pH 7, (b) aqueous alkali at pH 12, and (c) in concentrated mineral acid?

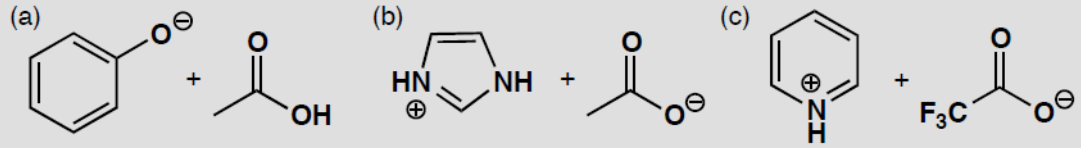


The CO_2H group will have a pKa of about 4–5 and the phenolic OH a pKa of about 10. So the carboxylic acid but not the phenol will be ionized at pH 7, they will both be ionized at pH 12, and there will be a mixture of free acid and protonated acid at very low pH. The proton will be on the carbonyl oxygen atom as this gives a delocalized cation.

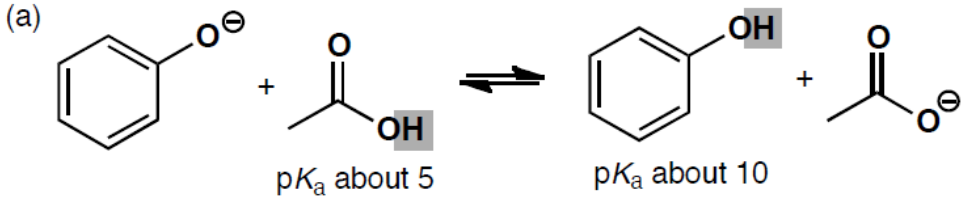


Problem 3

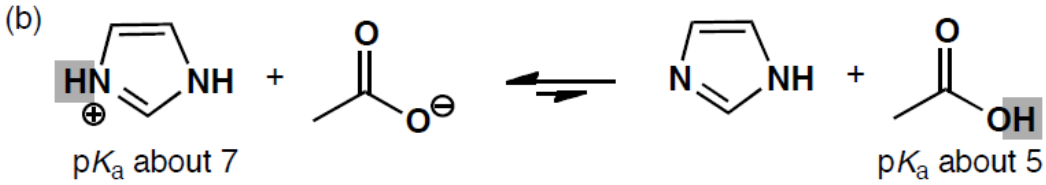
Suggest what species would be formed by each of these combinations of reagents. You are advised to use estimated pK_a values to help you and to beware of those cases where nothing happens.



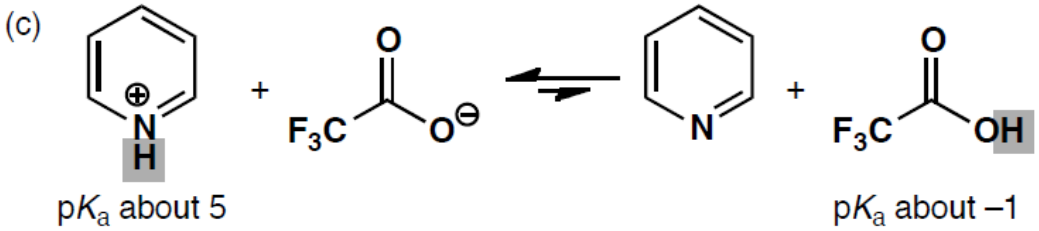
In each case one of the reagents might take a proton from the other. In example (a), would the phenolate anion remove a proton from acetic acid? The answer is yes because acetic acid is a much stronger acid than phenol The difference is five pH units so the equilibrium constant would be about 10^5 and the equilibrium would lie far across to the right.



Example (b) has a similar possible reaction but this time the pKa difference is much smaller and the other way so the equilibrium constant is 100 and favours the starting materials.

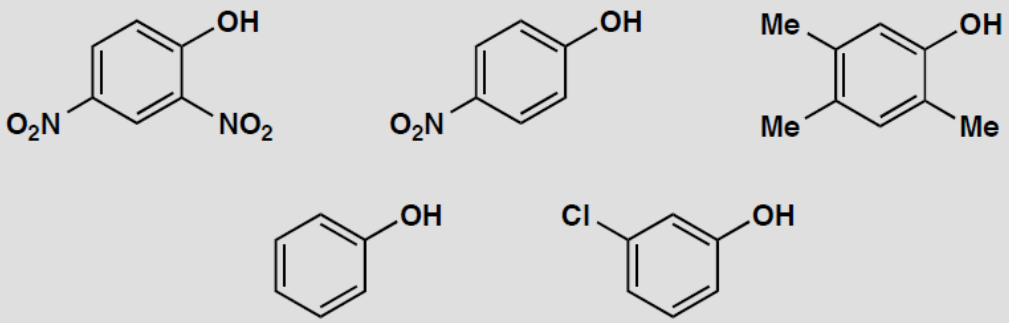


Example (c) is rather different. We do have another carboxylic acid but this is a much stronger acid because of the three fluorine atoms and the equilibrium is far over to the left.

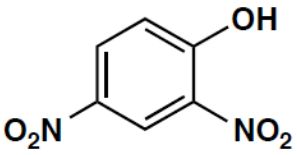


Problem 4

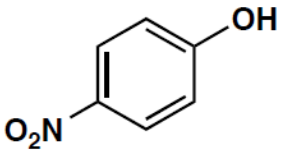
These phenols have approximate pK_a values of 4, 7, 9, 10, and 11. Suggest with explanations which pK_a value belongs to which phenol.



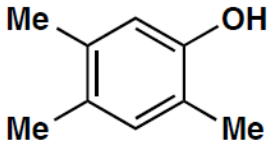
Electron-withdrawing effects make phenols more acidic and electrondonating effects make them less acidic. Phenol itself (the fourth example) has a pKa of 10. The only compound less acidic than phenol must be the third with three weakly electron-donating methyl groups. One chlorine atom has an inductive electron-withdrawing effect so the last compound has pKa 9. The remaining two have the powerful electron-withdrawing nitro group. So the first compound, with two nitro groups, must have pKa of 4 (making it as strong an acid as acetic acid) and the second, with one nitro group, must have pKa of 7.



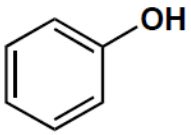
pK_a about 4



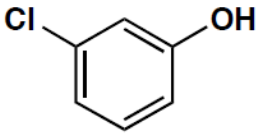
pK_a about 7



pK_a about 11



pK_a about 10



pK_a about 9

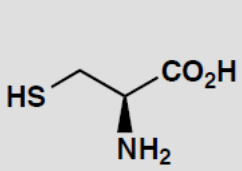
Problem 5

The pK_a values of these two amino acids are as follows:

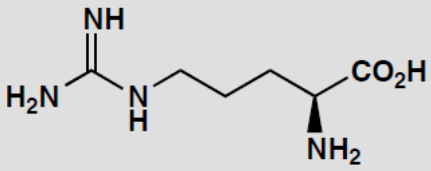
(a) cysteine: 1.8, 8.3, and 10.8

(b) arginine: 2.2, 9.0, and 13.2.

Assign these pK_a values to the functional groups in each amino acid and draw the most abundant structure that each molecule will have at pH 1, 7, 10, and 14.

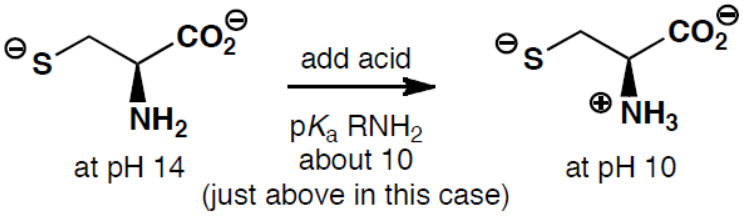


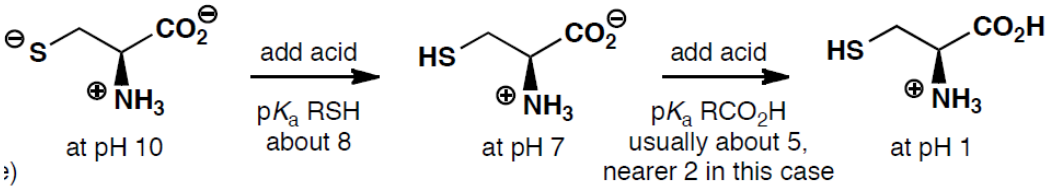
cysteine



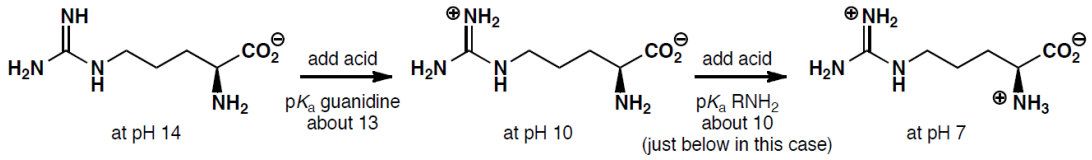
arginine

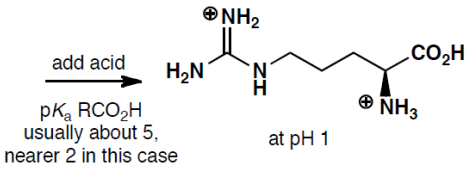
At high pH cysteine exists as a dianion as both the thiol and the carboxylic acid are anions. If we now add acid, at about pH 10 (actually 10.8) the amine will get protonated, then the thiol will be protonated at about pH 8 (actually 8.3) and finally the carboxylic acid will be protonated at low pH, rather lower than say MeCO2H, as the electron-withdrawing ammonium group increases its acidity (actually at 1.8).





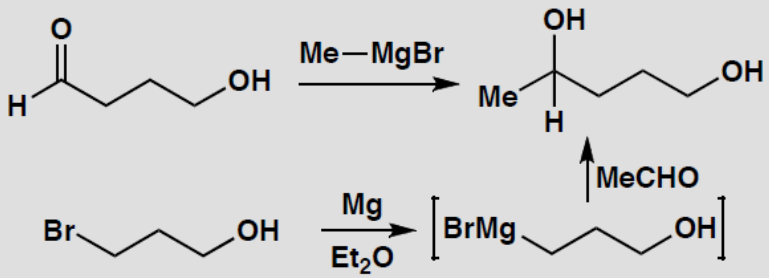
At high pH arginine exists as a monoanion—even the very basic guanidine group cannot be protonated at pH 14. If we now add acid, at about pH 13 the guanidine will get protonated, then the amino group will be protonated at about pH 10 (actually 9.0) and finally the carboxylic acid will be protonated at low pH. This carboxylic acid is rather more acidic than you might expect, but not surprisingly it is harder to protonate an anion in a molecule which already has an overall positive charge.



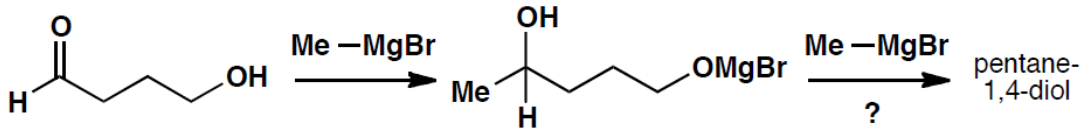


Problem 6

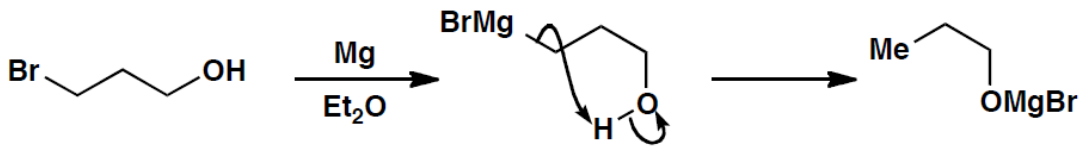
Neither of these two methods for making pentan-1,4-diol will work. What will happen instead?



The OH group is the Wicked Witch of the West in this problem. Whoever planned these syntheses expected it to lie quietly and do nothing. All chemists have to learn that things don't go our way just because we want them to do so. Here, although the OH group is only a weak acid (pKa about 16) it will give up its proton to the very basic Grignard reagents. In the first case, one molecule of Grignard is destroyed but the reaction might succeed if two equivalents were used.



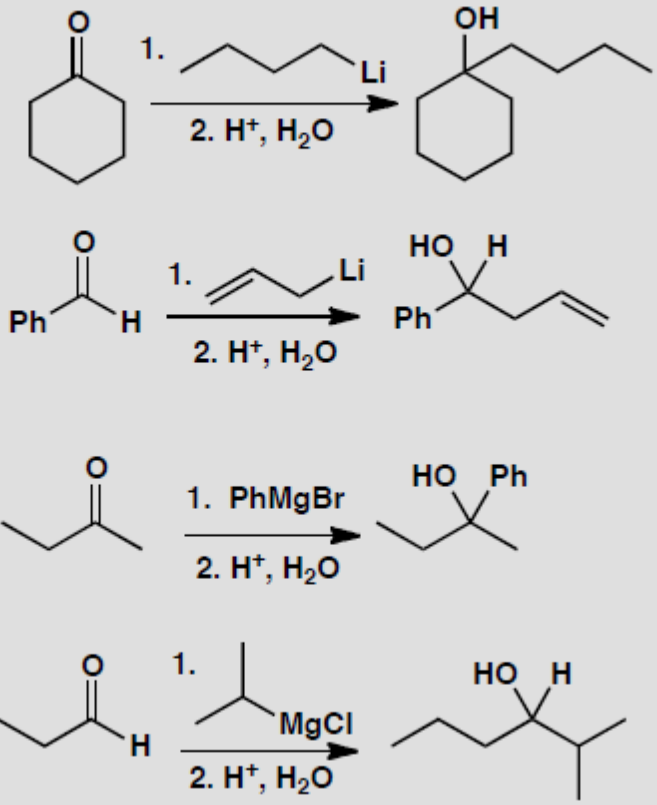
The second case is hopeless as the Grignard reagent destroys itself by intramolecular deprotonation. This synthesis could be rescued by putting a protecting group on the OH.



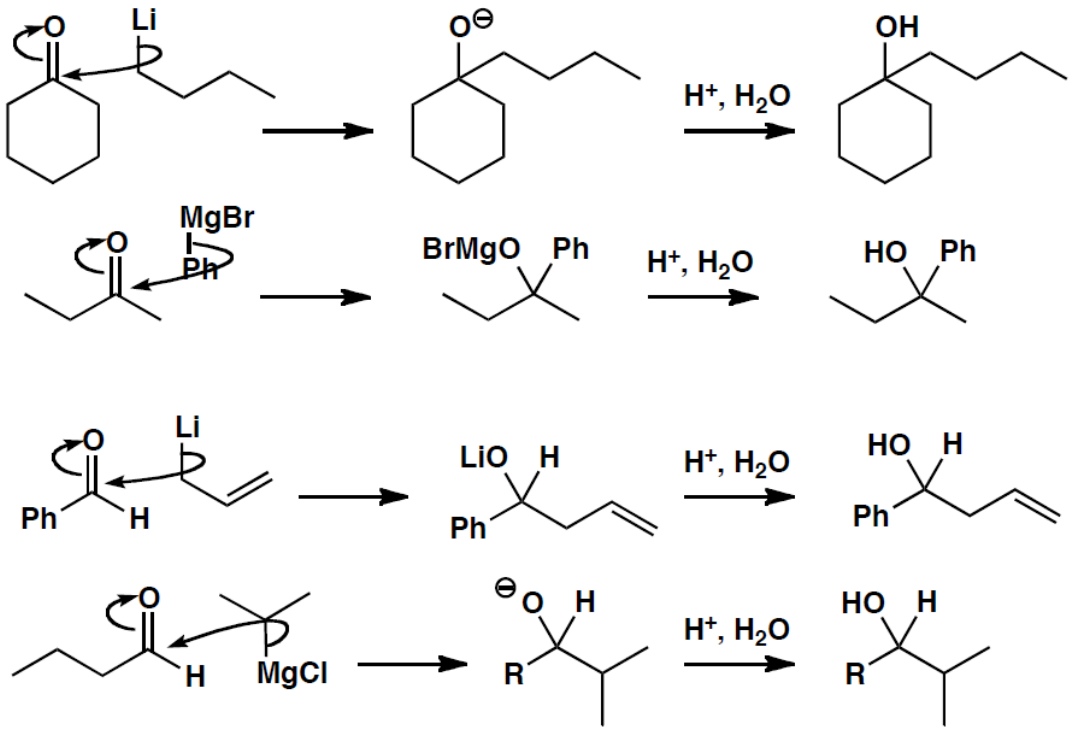
Tutorial 8

Problem 1

Propose mechanisms for the first four reactions in the chapter.

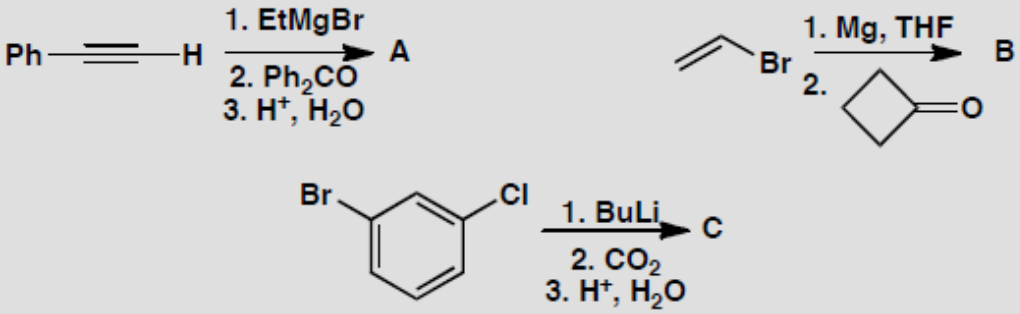


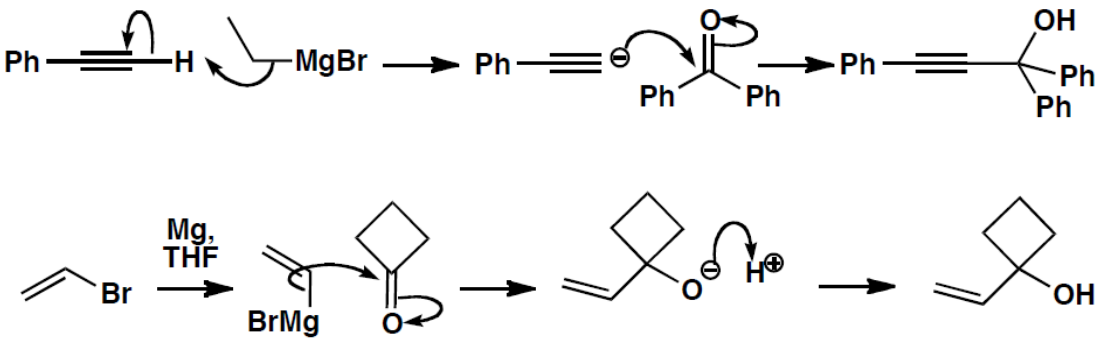
See lecture notes as well

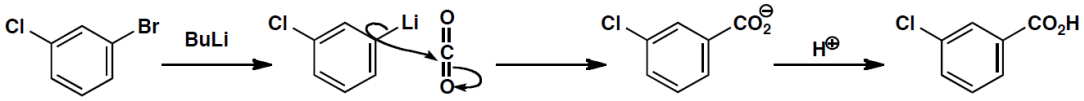


Problem 2

What products would be formed in these reactions?

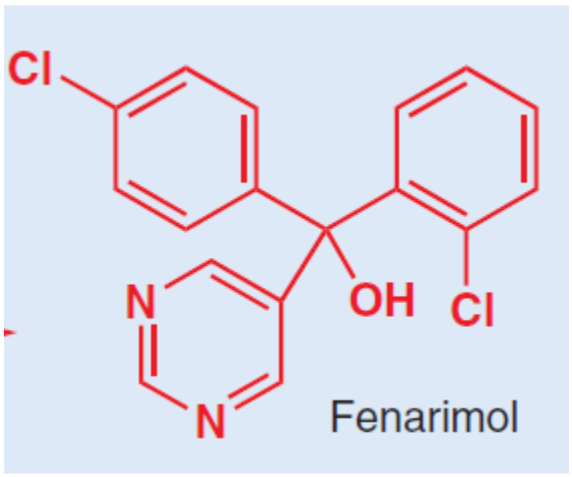


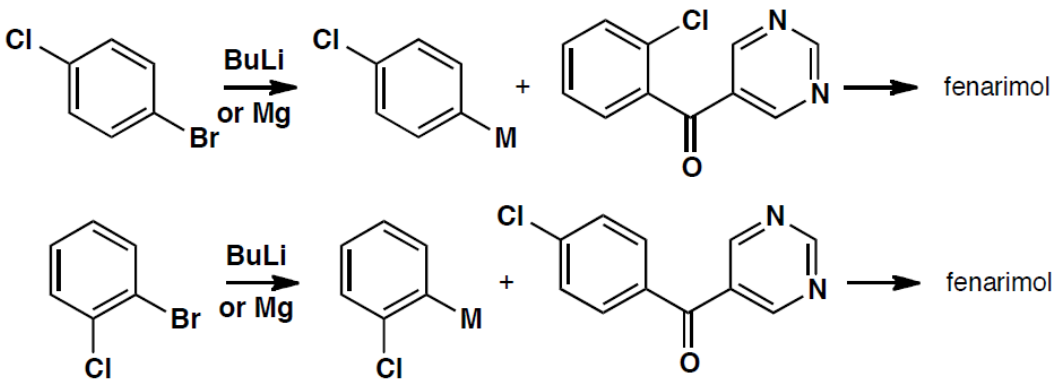




Problem 3

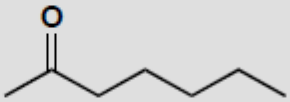
Suggest alternative routes to fenarimol different from the one in the textbook on p. 192. Remind yourself of the answer to problem 2 above.

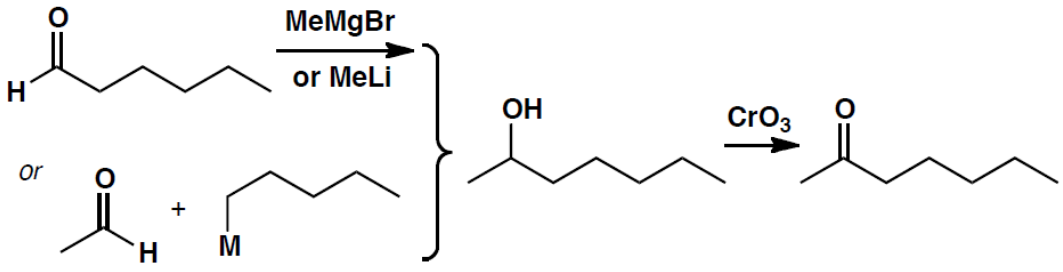




Problem 4

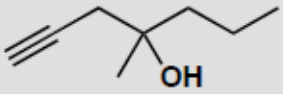
Suggest two syntheses of the bee pheromone heptan-2-one.





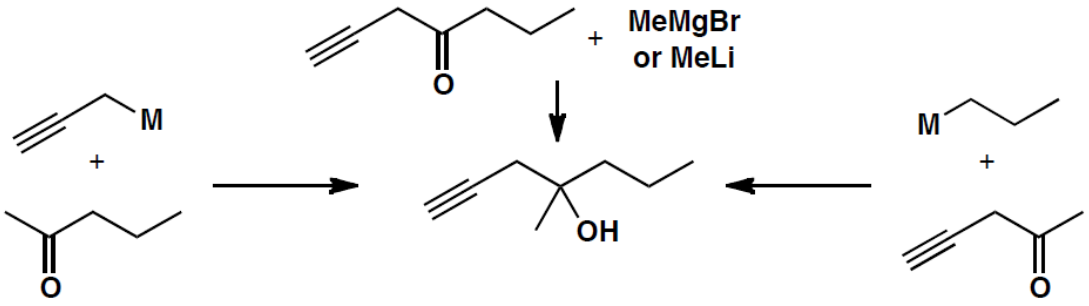
Problem 5

The synthesis of the gastric antisecretory drug rioprostil requires this alcohol.

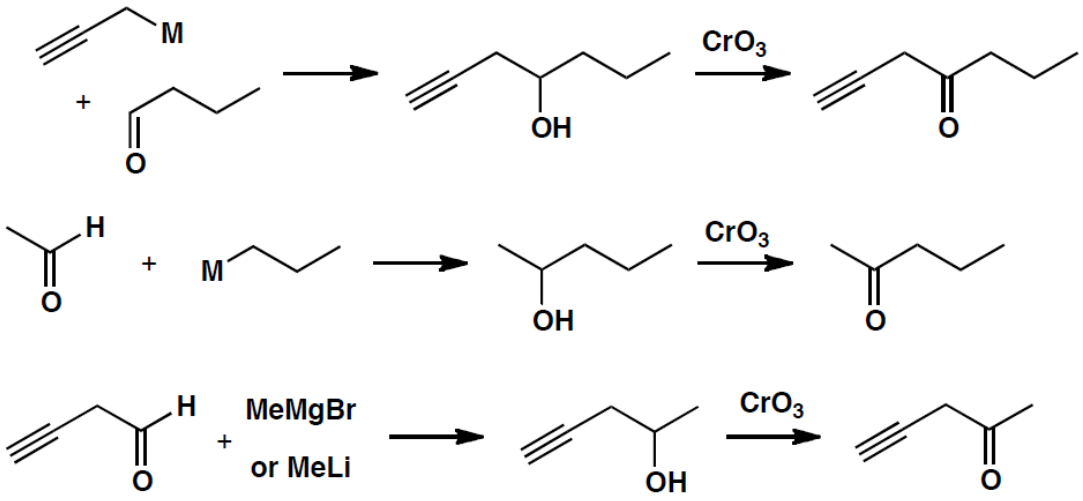


(a) Suggest possible syntheses starting from ketones and organometallics.
(b) Suggest possible syntheses of the ketones in part (a) from aldehydes and organometallics (don't forget about CrO_3 oxidation).

There are three one-step syntheses from ketones and organometallic compounds. We have used 'M' to indicate the metal—it might be Li or MgX (in other words, the organometallic could be an organolithium or a Grignard reagent).

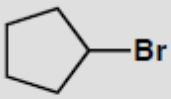


Each of these ketones can be made by oxidation of an alcohol that can in turn be made from an organometallic compound and an aldehyde.

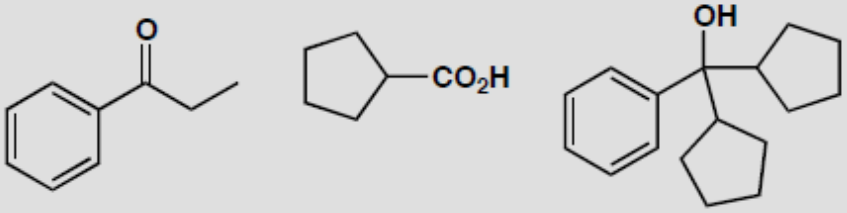


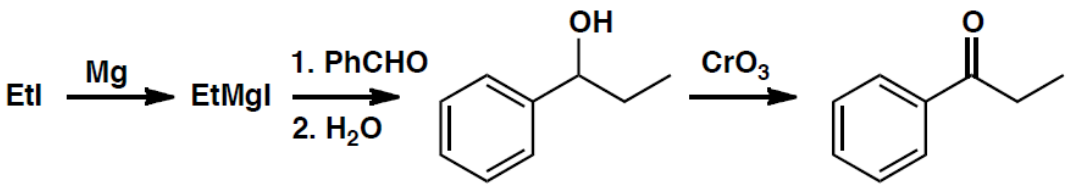
Problem 6

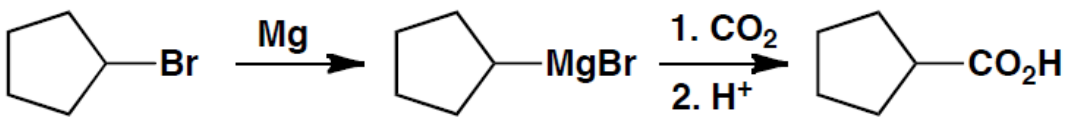
How could you use these four commercially available starting materials

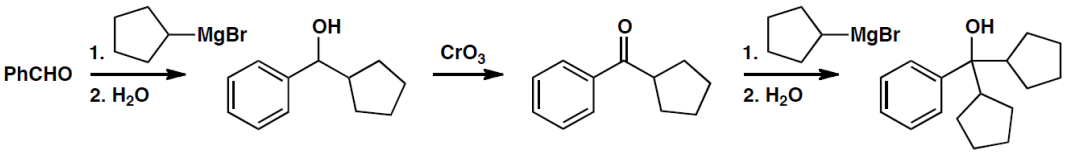
PhCHO EtI CO_2 

to make the following three compounds?





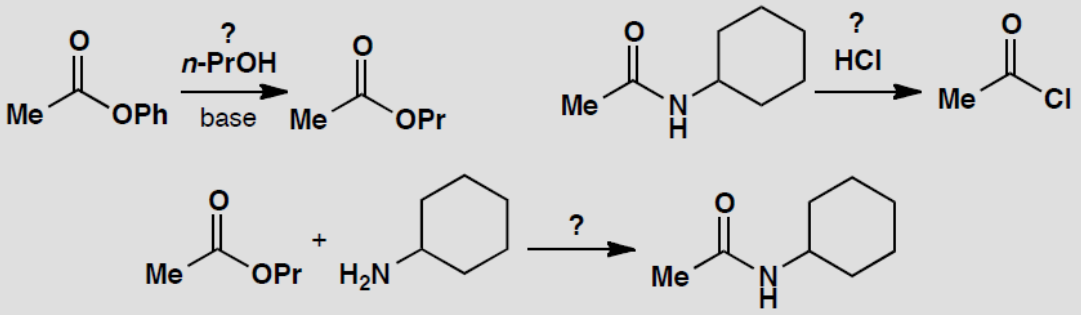


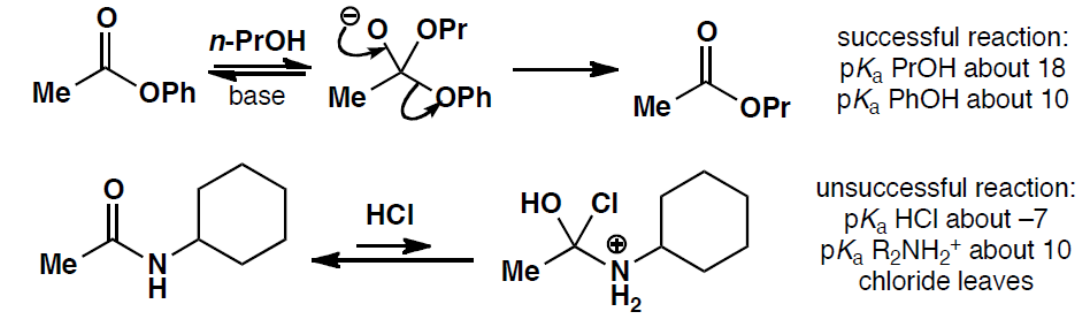


Tutorial 10

Problem 7

Predict the success or failure of these attempted substitutions at the carbonyl group. You should use estimated pK_a values in your answer and, of course, draw mechanisms.





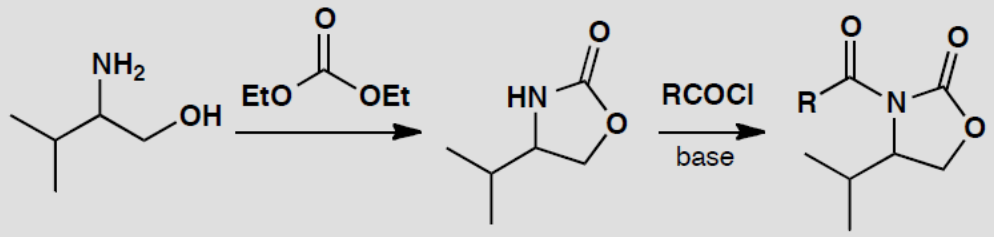
successful reaction:
pK_a PrOH about 18
pK_a PhOH about 10

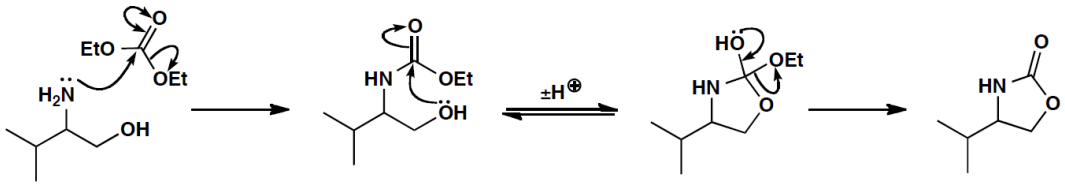
unsuccessful reaction:
pK_a HCl about –7
pK_a $R_2NH_2^+$ about 10
chloride leaves

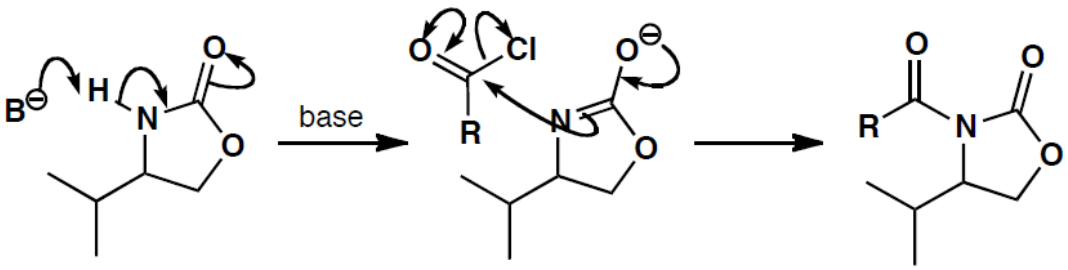
successful reaction:
pK_a PrOH about 18
pK_a R_2NH about 35
PrO^- leaves

Problem 8

Suggest mechanisms for these reactions.

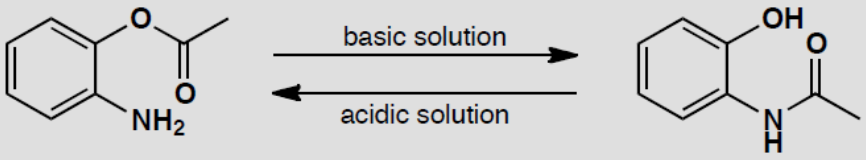


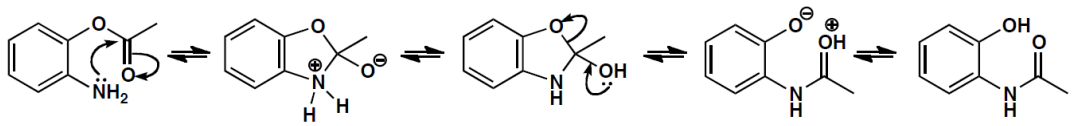




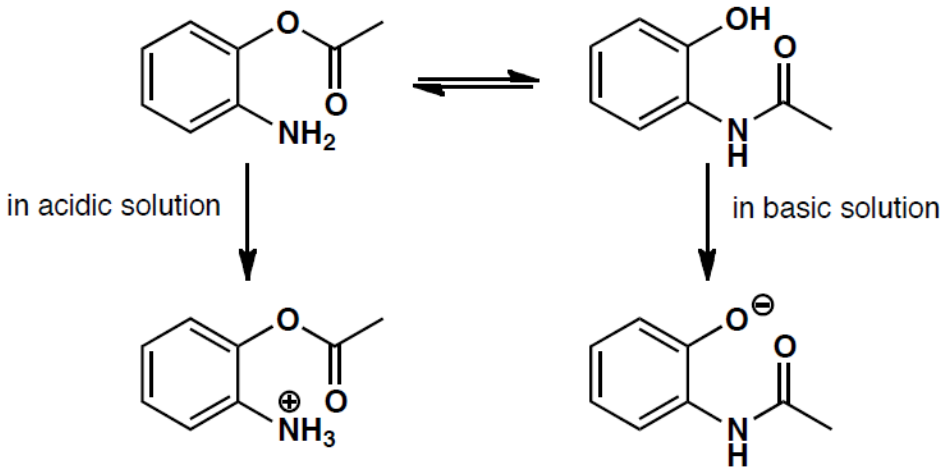
Problem 9

This reaction goes in one direction in acid solution and in the other direction in basic solution. Draw mechanisms for the reactions and explain why the product depends on the conditions.



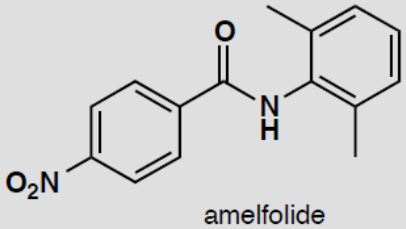


The amine attacks the ester in the usual way to give the tetrahedral intermediate which decomposes with the loss of the better leaving group: phenols are reasonably acidic (pKa PhOH = 10) so the phenoxy anion is a much better leaving group than ArNH–. In strongly basic solution, the phenol product is fully deprotonated, so again, the equilibrium lies to the right. In acidic solution the starting amine is fully protonated, pulling the equilibrium back over to the left.

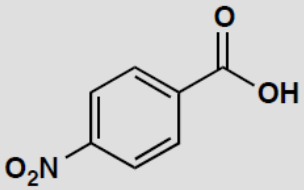


Problem 10

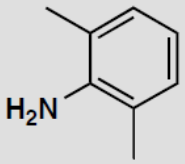
Amelfolide is a drug used to treat cardiac arrhythmia. Suggest how it could be made from 4-nitrobenzoic acid and 2,5-dimethylaniline.



amelfolide

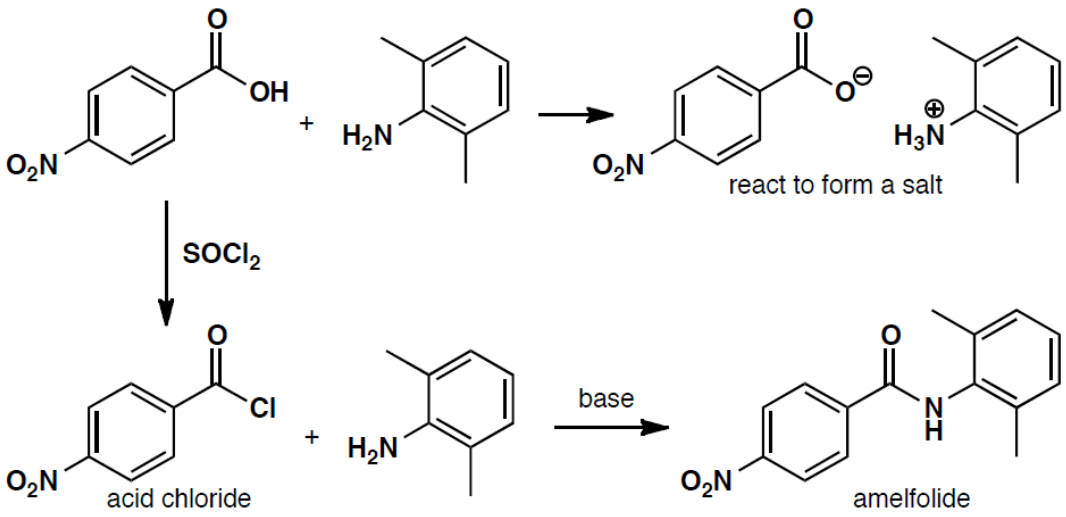


4-nitrobenzoic acid



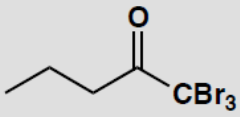
2,6-dimethylaniline

It is tempting to try and react the amine directly with the acid, but unfortunately the only product this would give is the ammonium carboxylate salt: the amine deprotonates the acid, and the carboxylate anion that results is no longer electrophilic. With alcohols, esters can be formed from carboxylic acids under acid catalysis, but with amines the acid catalyst just protonates the amine, and it is no longer nucleophilic! The simplest solution is to convert the carboxylic acid to an acid chloride and allow that to react with the amine. Additional base will neutralize the HCl by-product.

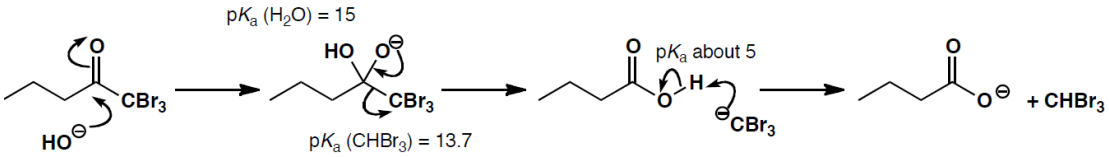


Problem 11

Given that the pK_a of tribromomethane, $CHBr_3$ (also known as bromoform) is 13.7, suggest what will happen when this ketone is treated with sodium hydroxide.



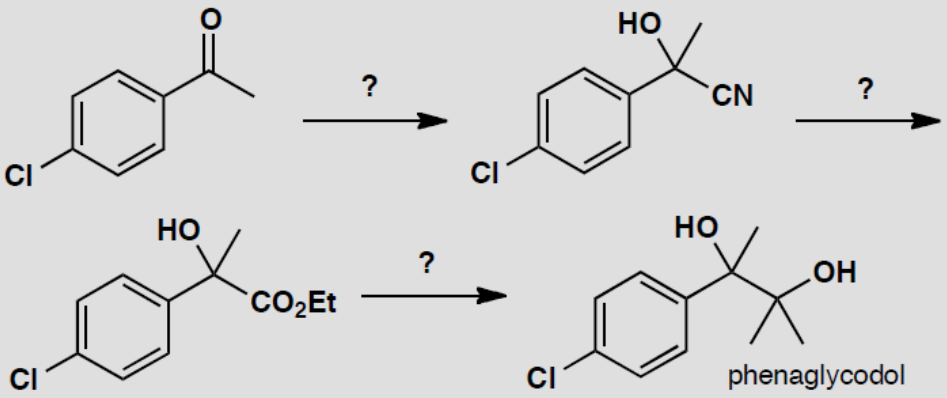
The best approach to new reactions is to start drawing curly arrows for steps you know are reasonable, and to see where they take you. Here, we are treating a carbonyl compound, an electrophile, with hydroxide, a nucleophile, so the first step is likely to be addition of hydroxide to the C=O group. You have seen many, many reactions that start this way.

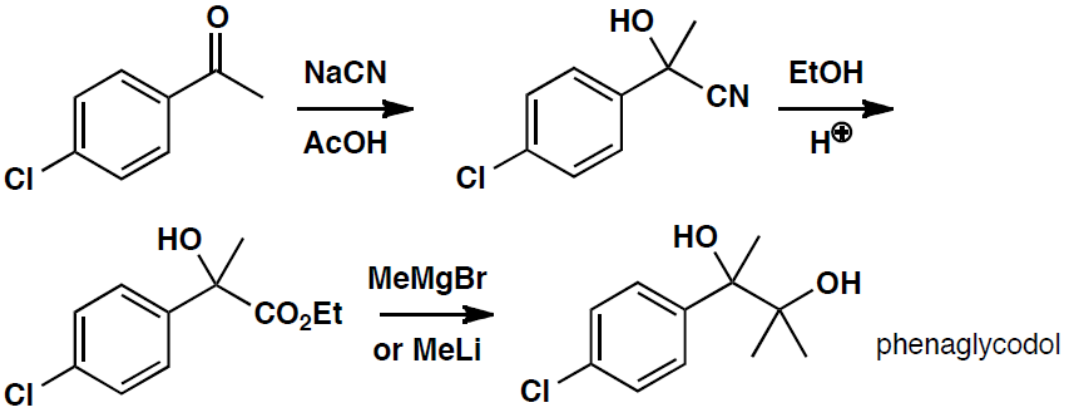


The result looks like a tetrahedral intermediate: the only possible leaving groups are the hydroxide (which takes us back to starting materials) or the anion Br_3C^-. You learnt in chapter 10 to use pKa to estimate leaving group ability, so the relatively low value of 13.7 should encourage you to eject it, giving a carboxylic acid as well. Neutralization of the acid by the tribromomethyl anion gives the products—the carboxylate anion and tribromomethane.

Problem 1

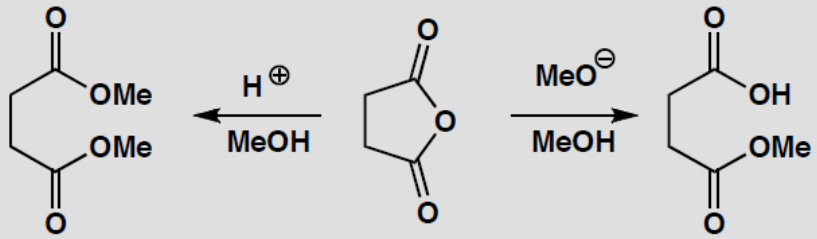
Suggest reagents to make the drug phenaglycodol by the route below.



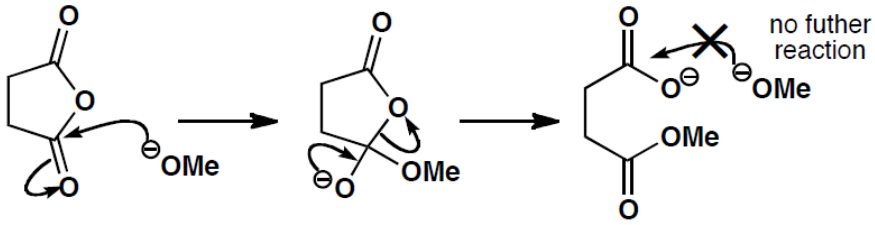


Problem 2

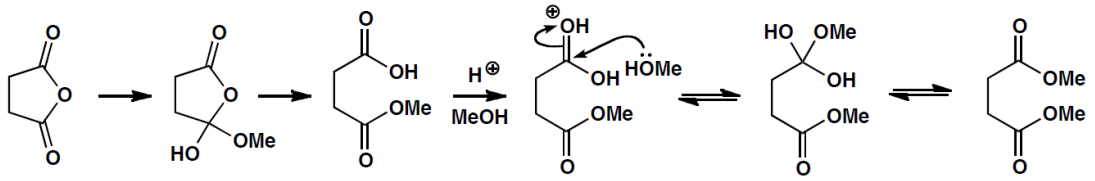
It is possible to make either the diester or the monoester of butanedioic acid (succinic acid) from the cyclic anhydride as shown. Why does one method give the diester and one the monoester?



In basic solution the nucleophile is methoxide ion. This strong nucleophile attacks the carbonyl group to give a tetrahedral intermediate having two possible leaving groups. The ester anion is preferred (pKa of RCO_2H about 5) to the alkoxide ion (pKa of ROH about 15). This carboxylate anion cannot be protonated in basic solution and is not attacked by methoxide ion.

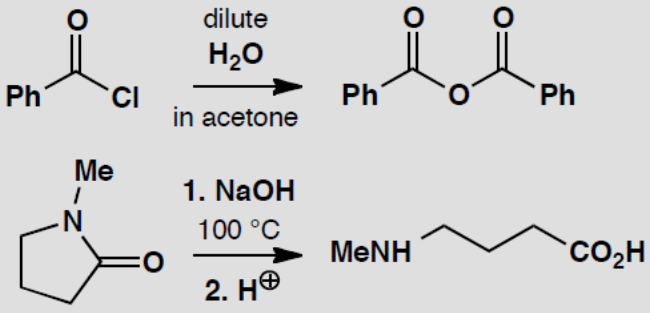


In acid solution the first reaction is similar, though the tetrahedral intermediate is neutral, and the carboxyl is still the better leaving group. The second esterification is now all right because methanol can attack the protonated carboxylic acid and water can be driven out after a second protonation. The second step is an equilibrium, with water and methanol about equal as leaving groups, but methanol is present in large excess as the solvent and drives the equilibrium across. We have omitted proton transfer steps.



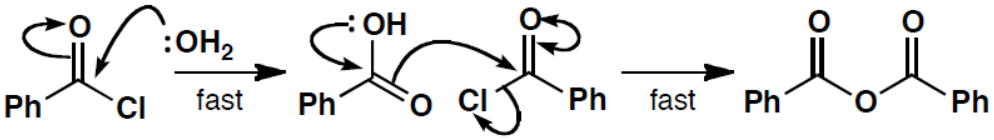
Problem 3

Suggest mechanisms for these reactions, explaining why these particular products are formed.

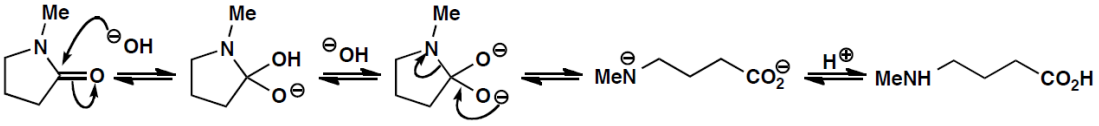


The acid chloride reacts rapidly with the water and the carboxylic acid produced reacts rapidly with a second molecule of acid chloride. The anhydride reacts much more slowly (pKa of HCl is –7 but the pKa of RCO_2H is about 5) with water so there is a good chance of stopping the reaction there, especially when we use a low concentration of water in acetone solution. In

this instance the chance is made a certainty because the anhydride precipitates from the solution and is no longer in equilibrium with the other reagents. It is usually possible to descend the reactivity sequence of acid derivatives.

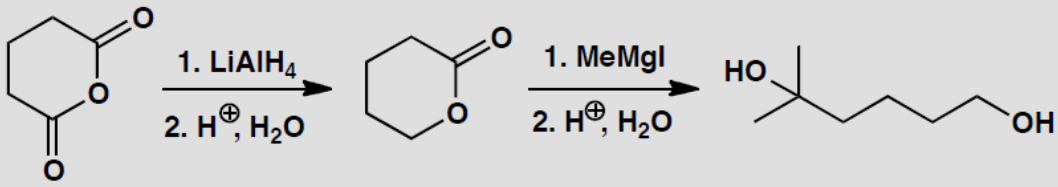


The second reaction is an example of the alkaline hydrolysis of amides. Though the nitrogen atom is never a good leaving group, it will leave from the dianion and, once gone, it is quickly protonated and does not come back. This example also benefits from the release of the slight strain in the five-membered ring.

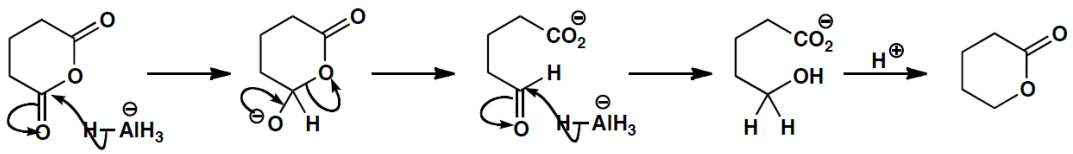


Problem 4

Give mechanisms for these reactions, explaining the selectivity (or lack of it!) in each case.

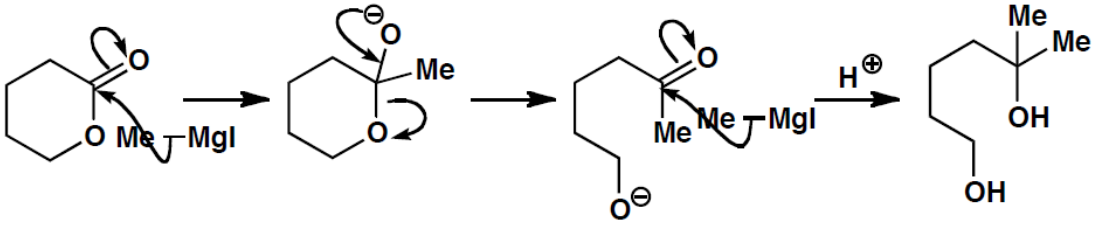


One of the carbonyl groups of the anhydride must be attacked by $LiAlH_4$ and we need to follow that reaction through to see what happens next. The first addition of AlH_4^- produces a tetrahedral intermediate that decomposes with the loss of the only possible leaving group, the carboxylate ion, to give an aldehyde. That too is quickly reduced by AlH_4^- to give the hydroxy-acid as its anion, which is resistant to further reduction. In the acidic aqueous work-up, excess LiAlH4 is instantly destroyed and the hydroxy-acid cyclizes to the lactone. The fact that the lactone is not formed under the reaction conditions is important: if it were, then it too would be reduced by the $LiAlH_4$.



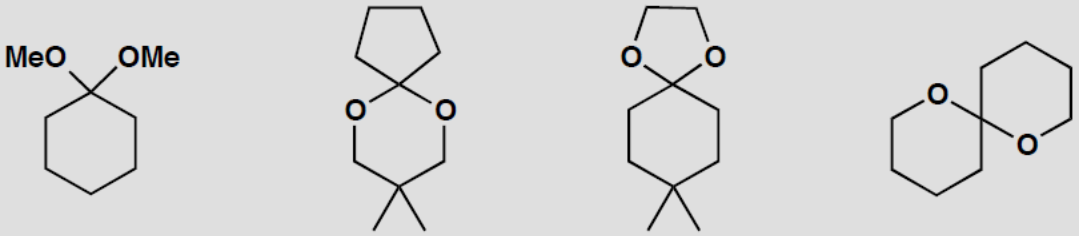
The second reaction starts similarly with the Grignard reagent adding to the ester carbonyl group and the tetrahedral intermediate losing the only possible leaving group. Again, a

reactive carbonyl compound is produced: a ketone that is more electrophilic than the ester, so it adds the Grignard reagent even faster. Work-up in aqueous acid gives the diol.

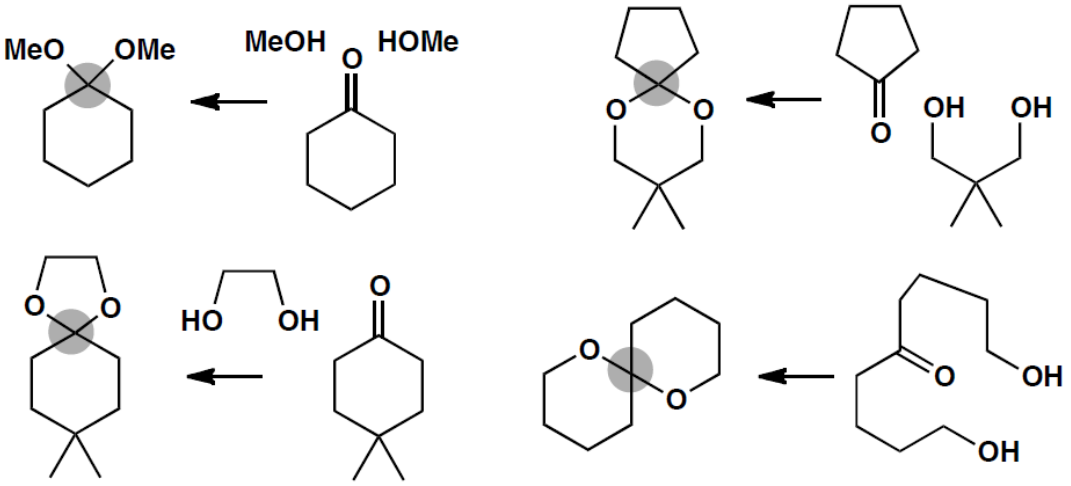


Problem 6

Each of these compounds is an acetal, that is a molecule made from an aldehyde or ketone and two alcohol groups. Which compounds were used to make these acetals?

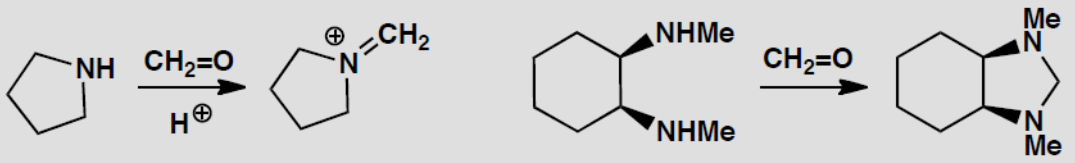


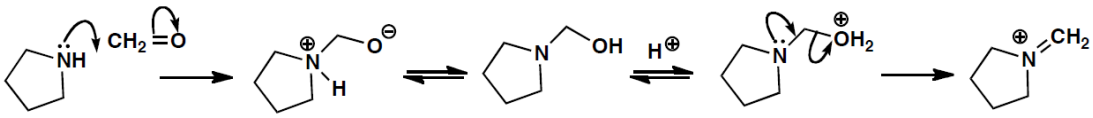
All we have to do is to identify the hidden carbonyl group by finding the only carbon atom having two C–O bonds. This atom is marked with a grey circle. If you imagine breaking the two C–O bonds you will discover the carbonyl group and the alcohols.

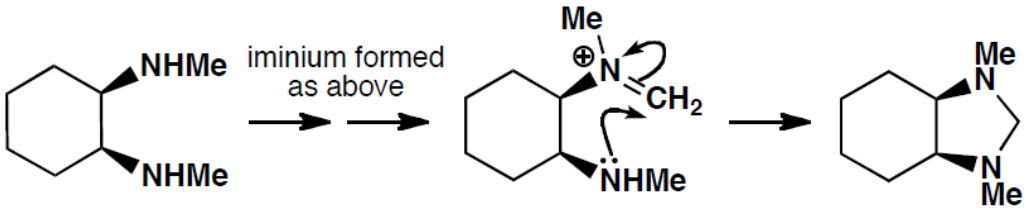


Problem 7

Suggest mechanisms for these two reactions of the smallest aldehyde, formaldehyde (methanal $CH_2=O$).

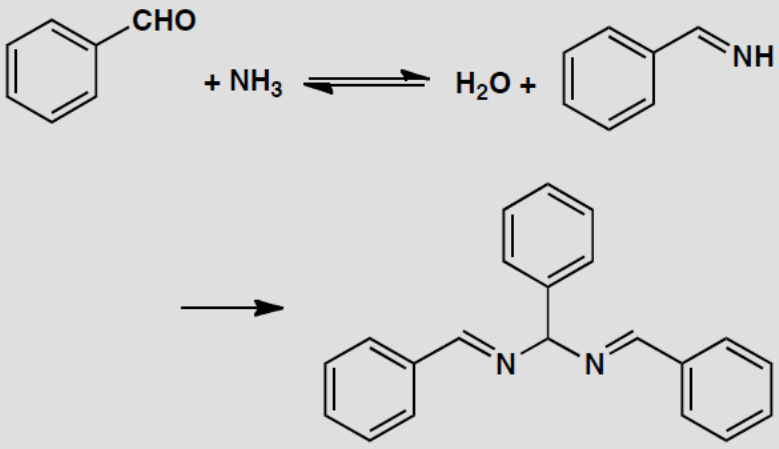




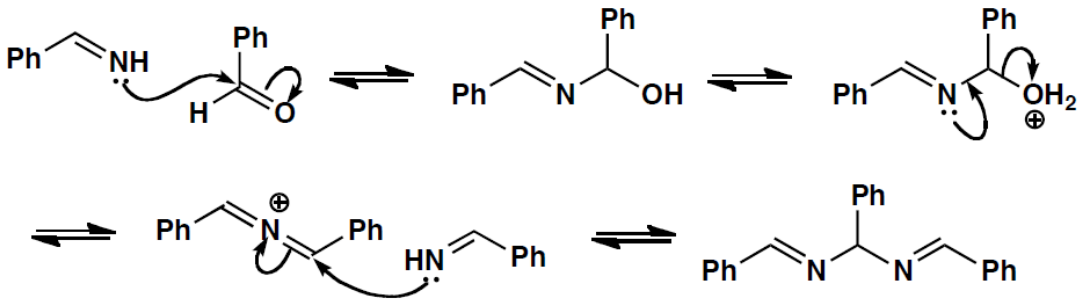


Problem 8

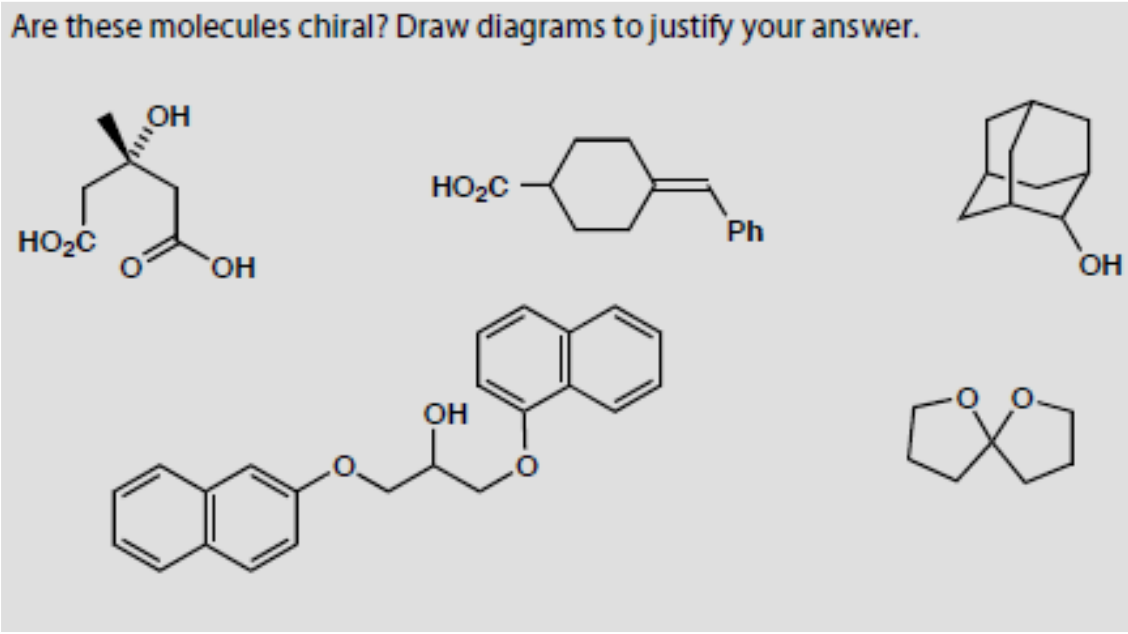
This stable product can be isolated from the reaction between benzaldehyde and ammonia. Suggest a mechanism.



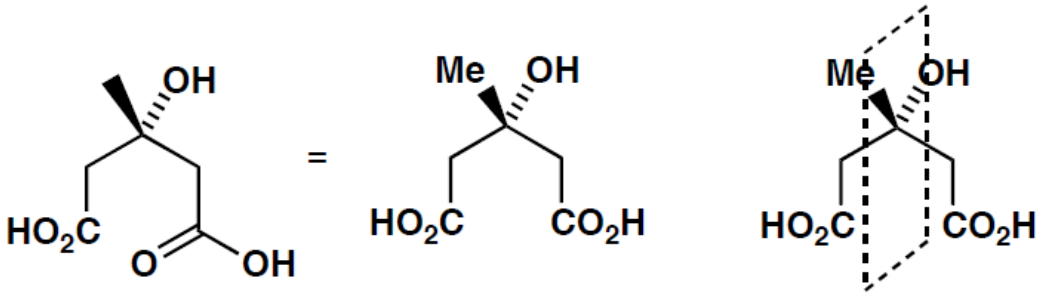
Imine formation follows the usual pathway (see problem 7 above) but this imine is unstable, as are most primary imines, and it reacts with more benzaldehyde. This reaction starts normally enough but dehydration of the first intermediate produces a strange looking cation with two double bonds to the same nitrogen atom. Addition of another imine gives the final product. The benzene rings play no part in these reactions so we shall represent them as Ph, but they do stablize the final product by conjugation with the imines.



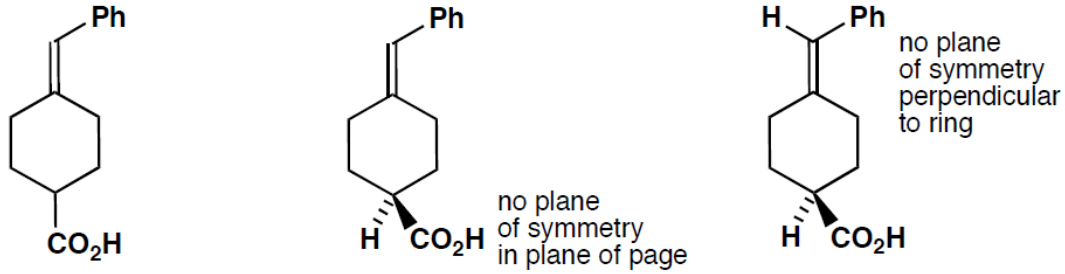
Problem 9



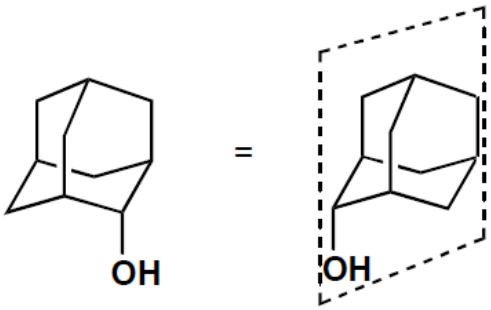
Only one thing matters: does the molecule have a plane of symmetry? We need to redraw some of them to see if they do. On no account look for chiral centres or carbon atoms with four different groups or anything else. Just look for a plane of symmetry. If the molecule has one, it isn't chiral. The first compound has been drawn with carboxylic acids represented in two different ways. The two CO2H groups are in fact the same and the molecule has a plane of symmetry (shown by the dashed lines). It isn't chiral.



The second compound is chiral but if you got this wrong don't be dismayed. Making a model would help but there are only two plausible candidate planes of symmetry: the ring itself, in the plane of the page, and a plane at right angles to the ring. The molecule redrawn below with the tetrahedral centre displayed shows that the plane of the page isn't a plane of symmetry as the CO_2H is on one side and the H on the other, and neither is the plane perpendicular to the ring, as Ph is on one side and H on the other. No plane of symmetry: molecule is chiral.

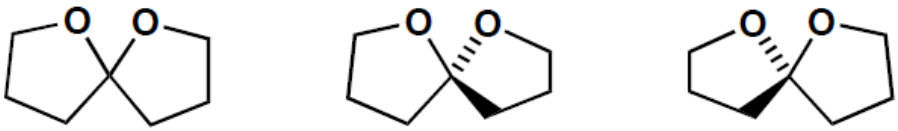


The third compound is not chiral because of its high symmetry. All the CH_2 groups are identical so the alcohol can be attached to any of them. The plane of symmetry (shown by the dotted lines) may be easier to see after redrawing, and will certainly be much easier to see if you make a model.



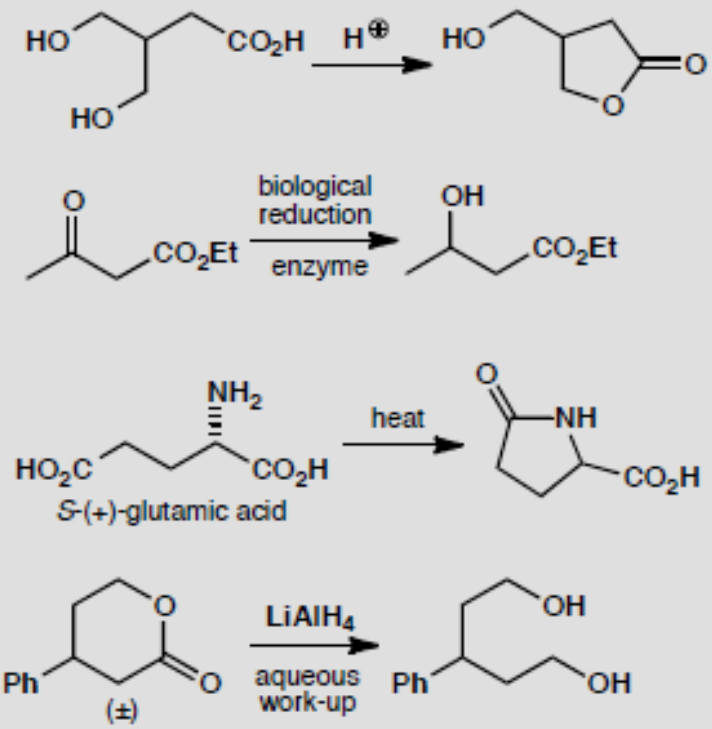
The fourth compound is chiral as the naphthalene rings have a different substitution. (textbook model answer is wrong as the molecule in the question is different than the model answer)

The final acetal (which is a spiro compound) is drawn flat but the central carbon atom must in fact be tetrahedral so that the two rings are orthogonal. By drawing first one and then the other ring in the plane of the page it is easy to see that neither ring is a plane of symmetry for the other because of the oxygen atoms.



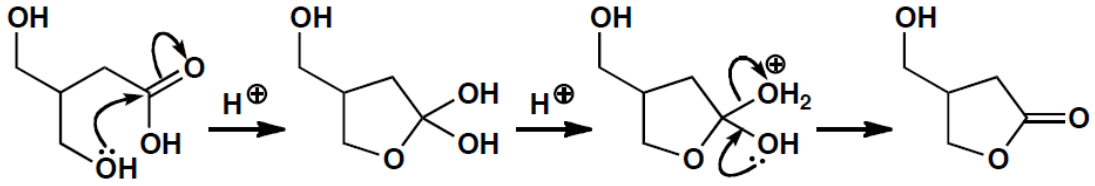
Problem 10

In each case state, with explanations, whether the products of these reactions are chiral and/or enantiomericaly pure.

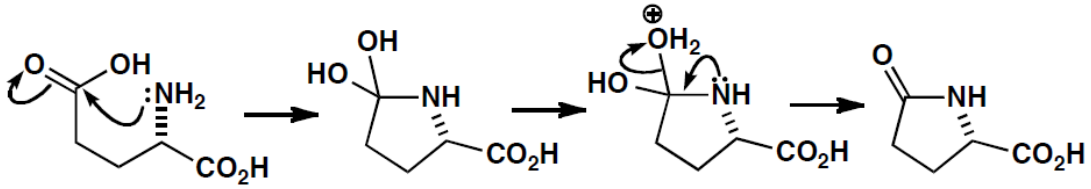


We need a mechanism for each reaction, a stereochemical description for each starting material (achiral, chiral? enantiomerically enriched?) and an analysis of what happens to the stereochemistry in each reaction. Don't forget: you can't get single enantiomers out of nothing—if everything that goes into a reaction is racemic or achiral, so is the product.

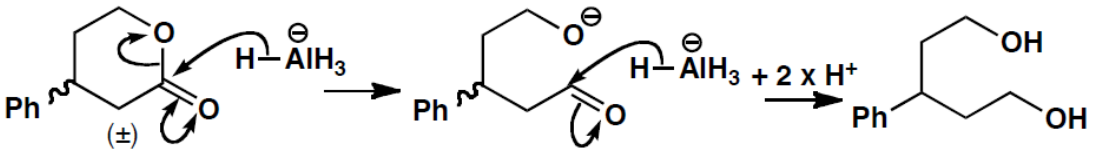
In the first reaction the starting material is achiral as the two CH_2OH side chains are identical. The product is chiral as it has no plane of symmetry but it cannot be one enantiomer as that would require one of the CH_2OH side chains to cyclize rather than the other. It must be racemic.



The starting material for the second reaction is planar and achiral. If the reagent had been sodium borohydride, the product would be chiral but racemic. But an enzyme, because it is made up of enantiomerically pure components (amino acids), can deliver hydride to one side of the ketone only. We expect the product to be enantiomerically enriched. In the third reaction, the starting material is one enantiomer of a chiral compound. So we need to ask what happens to the chiral centre during the reaction. The answer is nothing as the reaction takes place between the amine and the carboxylic acid. The product is a single enantiomer too.

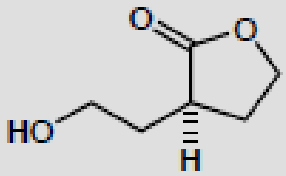


The final problem is a bit of a trick. The starting material is chiral, but racemic while the product is achiral as the two CH_2CH_2OH side chains are identical so there can be a plane of symmetry between them. The mechanism doesn't really matter but we might as well draw it.



Problem 11

This compound racemizes in base. Why is that?



Ester exchange in base goes in this case through a symmetrical (achiral) tetrahedral intermediate with a plane of symmetry. Loss of the right hand leaving group gives one enantiomer of the ester and loss of the left hand leaving group gives the other.

